

Experiment No. 9

1. Find instructors who have not taught any course.

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```
SELECT name
FROM instructor
WHERE ID NOT IN (
    SELECT DISTINCT ID FROM teaches
);
```

Results Explain Describe Saved SQL History

NAME

Turing

2. View: section_info combining section and course details

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```
CREATE VIEW section_info AS
SELECT S.course_id, C.title, S.semester, S.year, S.building, S.room_number
FROM section S
JOIN course C ON S.course_id = C.course_id;
```

Results Explain Describe Saved SQL History

COURSE_ID	TITLE	SEMESTER	YEAR	BUILDING	ROOM_NUMBER
BIO-101	Intro to Biology	Fall	2009	Watson	101
BIO-101	Intro to Biology	Spring	2010	Watson	101
BIO-201	Genetics	Spring	2010	Watson	102
PHY-101	Intro to Physics	Fall	2009	Newton	202
PHY-201	Quantum Mechanics	Spring	2010	Newton	203
CS-101	Intro to CS	Spring	2010	Taylor	303
CS-201	Algorithms	Fall	2009	Taylor	304
MATH-101	Calculus I	Fall	2009	Euler	201
CHEM-101	Organic Chemistry	Spring	2010	Curie	501

3. Courses not taken by any student.

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```
SELECT course_id
FROM course
WHERE course_id NOT IN (
    SELECT DISTINCT course_id FROM takes
);
```

Results Explain Describe Saved SQL History

COURSE_ID

PHY-201

4. Students who have only taken Computer Science courses.

```
SELECT S.name
FROM student S
WHERE NOT EXISTS (
    SELECT *
    FROM takes T JOIN course C ON T.course_id = C.course_id
    WHERE T.ID = S.ID AND C.dept_name <> 'Computer Science'
);
```

Results Explain Describe Saved SQL History

NAME

Frank

Charlie

5. Departments where no instructor earns more than 60000.

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```
SELECT dept_name
FROM department
WHERE dept_name NOT IN (
    SELECT dept_name
    FROM instructor
    WHERE salary > 60000
);
```

Results Explain Describe Saved SQL History

DEPT_NAME
Biology
Chemistry
Mathematics

6. Students who have taken more than one course in Fall 2009.

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☒ Autocommit Display 10 Save Run

```
SELECT ID
FROM takes
WHERE semester = 'Fall' AND year = 2009
GROUP BY ID
HAVING COUNT(*) > 1;
```

Results Explain Describe Saved SQL History

no data found

7. Instructors who taught all sections of a specific course.

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☒ Autocommit Display 10 Save Run

```
SELECT ID
FROM teaches
WHERE course_id = 'BIO-101'
GROUP BY ID
HAVING COUNT(*) = (
    SELECT COUNT(*) FROM section WHERE course_id = 'BIO-101'
);
```

Results Explain Describe Saved SQL History

ID
12121

1 rows returned in 0.00 seconds CSV Export

8. Courses with no instructors assigned.

☒ Autocommit Display 10 Save Run

```
SELECT course_id
FROM course
WHERE course_id NOT IN (
    SELECT DISTINCT course_id FROM teaches
);
```

Results Explain Describe Saved SQL History

COURSE_ID
CHEM-101

9. Students who have the same department as instructor 'Mozart'.

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```
SELECT name
FROM student
WHERE dept_name = (
  SELECT dept_name FROM instructor WHERE name = 'Mozart'
);
```

Results Explain Describe Saved SQL History

NAME
Bob

10. Names of instructors who have taught a course taken by student 'Charlie'.

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT I.name
FROM instructor I
WHERE I.ID IN (
  SELECT T1.ID
  FROM teaches T1
  WHERE (T1.course_id, T1.sec_id, T1.semester, T1.year) IN (
    SELECT T2.course_id, T2.sec_id, T2.semester, T2.year
    FROM takes T2 JOIN student S ON T2.ID = S.ID
    WHERE S.name = 'Charlie'
  ));
```

Results Explain Describe Saved SQL History

NAME
Srinivasan

11. List departments and number of instructors in each.

☒ Autocommit Display 10 Save Run

```
SELECT dept_name, COUNT(*) AS num_instructors
FROM instructor
GROUP BY dept_name;
```

Results Explain Describe Saved SQL History

DEPT_NAME	NUM_INSTRUCTORS
Mathematics	1
Physics	2
Computer Science	2
Biology	2

12. Average salary by department, ordered descending.

☒ Autocommit Display 10 Save Run

```
SELECT dept_name, AVG(salary) AS avg_sal
FROM instructor
GROUP BY dept_name
ORDER BY avg_sal DESC;
```

Results Explain Describe Saved SQL History

DEPT_NAME	AVG_SAL
Computer Science	72500
Physics	56000
Biology	55000
Mathematics	43000

13. Departments where total instructor salary exceeds 100000.

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```
SELECT dept_name
FROM instructor
GROUP BY dept_name
HAVING SUM(salary) > 100000;
```

Results Explain Describe Saved SQL History

DEPT_NAME
Physics
Computer Science
Biology

14. Students with total credits > department average.

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```
WITH dept_avg AS (
  SELECT dept_name, AVG(tot_cred) AS avg_cred
  FROM student
  GROUP BY dept_name
)
SELECT S.ID, S.name
FROM student S
JOIN dept_avg D ON S.dept_name = D.dept_name
WHERE S.tot_cred > D.avg_cred;
```

Results Explain Describe Saved SQL History

ID	NAME
1003	Charlie
1007	Grace

15. Total courses taught per instructor.

☒ Autocommit Display 10 Save Run

```
SELECT ID, COUNT(DISTINCT course_id) AS courses_taught
FROM teaches
GROUP BY ID;
```

Results Explain Describe Saved SQL History

ID	COURSES_TAUGHT
10101	2
12121	2
15151	1
22222	1
32343	1
98989	1

Query returned in 0.00 seconds

16. Course with highest number of students.

☒ Autocommit Display 10 Save Run

```
SELECT course_id
FROM takes
GROUP BY course_id
HAVING COUNT(DISTINCT ID) >= ALL (
  SELECT COUNT(DISTINCT ID)
  FROM takes
  GROUP BY course_id
);
```

Results Explain Describe Saved SQL History

COURSE_ID
BIO-101

19. Departments with most offered courses.

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☒ Autocommit Display 10 Save Run

```
SELECT dept_name
FROM course
GROUP BY dept_name
HAVING COUNT(*) >= ALL (
  SELECT COUNT(*) FROM course GROUP BY dept_name
);
```

Results Explain Describe Saved SQL History

DEPT_NAME
Physics
Computer Science
Biology

20. Instructors who have taught in multiple semesters.

☒ Autocommit Display 10 Save Run

```
SELECT ID
FROM teaches
GROUP BY ID
HAVING COUNT(DISTINCT semester || year) > 1;
```

Results Explain Describe Saved SQL History

ID
10101
12121

21. Total students per department.

☒ Autocommit Display 10 Save Run

```
SELECT dept_name, COUNT(*) AS num_students
FROM student
GROUP BY dept_name;
```

Results Explain Describe Saved SQL History

DEPT_NAME	NUM_STUDENTS
Physics	1
Mathematics	1
Chemistry	1
Computer Science	2
Biology	2

22. View: instructor_salary with ID, name, salary.

☒ Autocommit Display 10 Save Run

```
SELECT ID, name, salary
FROM instructor;
```

Results Explain Describe Saved SQL History

ID	NAME	SALARY
32343	Kline	43000
98989	Crick	60000
76766	Turing	80000
10101	Srinivasan	65000
12121	Wu	50000
15151	Mozart	40000
22222	Einstein	72000

24.View: course_details with title, dept_name, instructor_name

☒ Autocommit Display 10 Save Run

```
SELECT C.title, C.dept_name, I.name AS instructor
FROM course C
JOIN teaches T ON C.course_id = T.course_id
JOIN instructor I ON T.ID = I.ID;
```

Results Explain Describe Saved SQL History

TITLE	DEPT_NAME	INSTRUCTOR
Intro to Biology	Biology	Wu
Intro to Biology	Biology	Wu
Genetics	Biology	Wu
Genetics	Biology	Crick
Intro to Physics	Physics	Mozart
Quantum Mechanics	Physics	Einstein
Intro to CS	Computer Science	Srinivasan
Algorithms	Computer Science	Srinivasan
Calculus I	Mathematics	Kline

25.Retrieve course_details

☒ Autocommit Display 10 Save Run

```
SELECT *
FROM course;
```

Results Explain Describe Saved SQL History

COURSE_ID	TITLE	DEPT_NAME	CREDITS
BIO-101	Intro to Biology	Biology	4
BIO-201	Genetics	Biology	4
PHY-101	Intro to Physics	Physics	3
PHY-201	Quantum Mechanics	Physics	4
CS-101	Intro to CS	Computer Science	3
CS-201	Algorithms	Computer Science	3
MATH-101	Calculus I	Mathematics	3
CHEM-101	Organic Chemistry	Chemistry	4

26.View: student_enrollment showing student name and courses

☒ Autocommit Display 10 Save Run

```
SELECT S.name AS student, C.title AS course
FROM student S
JOIN takes T ON S.ID = T.ID
JOIN course C ON T.course_id = C.course_id;
```

Results Explain Describe Saved SQL History

STUDENT	COURSE
Grace	Intro to Biology
Alice	Intro to Biology
Alice	Intro to Biology
Grace	Genetics
Bob	Intro to Physics
Charlie	Intro to CS
Frank	Algorithms
Daisy	Calculus I
Eve	Organic Chemistry

9 rows returned in 0.00 seconds [CSV Export](#)

27.View: active_courses showing all currently taught courses

☒ Autocommit Display 10 Save Run

```
CREATE VIEW active_courses AS
SELECT DISTINCT course_id
FROM teaches;
```

Results Explain Describe Saved SQL History

COURSE_ID
CS-101
BIO-201
MATH-101
BIO-101
PHY-101
CS-201
PHY-201

28.View: department_stats with number of instructors and students

☒ Autocommit Display 10 Save Run

```
CREATE VIEW department_stats AS
SELECT D.dept_name,
       (SELECT COUNT(*) FROM instructor I WHERE I.dept_name = D.dept_name) AS instructors,
       (SELECT COUNT(*) FROM student S WHERE S.dept_name = D.dept_name) AS students
FROM department D;
```

Results Explain Describe Saved SQL History

DEPT_NAME	INSTRUCTORS	STUDENTS
Biology	2	2
Chemistry	0	1
Computer Science	2	2
Mathematics	1	1
Physics	2	1

29.

students who have taken a course taught by 'Wu'.

☒ Autocommit Display 10 Save Run

```
SELECT name
FROM student
WHERE ID IN (
  SELECT T.ID
  FROM takes T
  WHERE (T.course_id, T.sec_id, T.semester, T.year) IN (
    SELECT course_id, sec_id, semester, year
    FROM teaches
    WHERE ID = (
      SELECT ID FROM instructor WHERE name = 'Wu')
  )
);
```

Results Explain Describe Saved SQL History

NAME
Alice
Grace